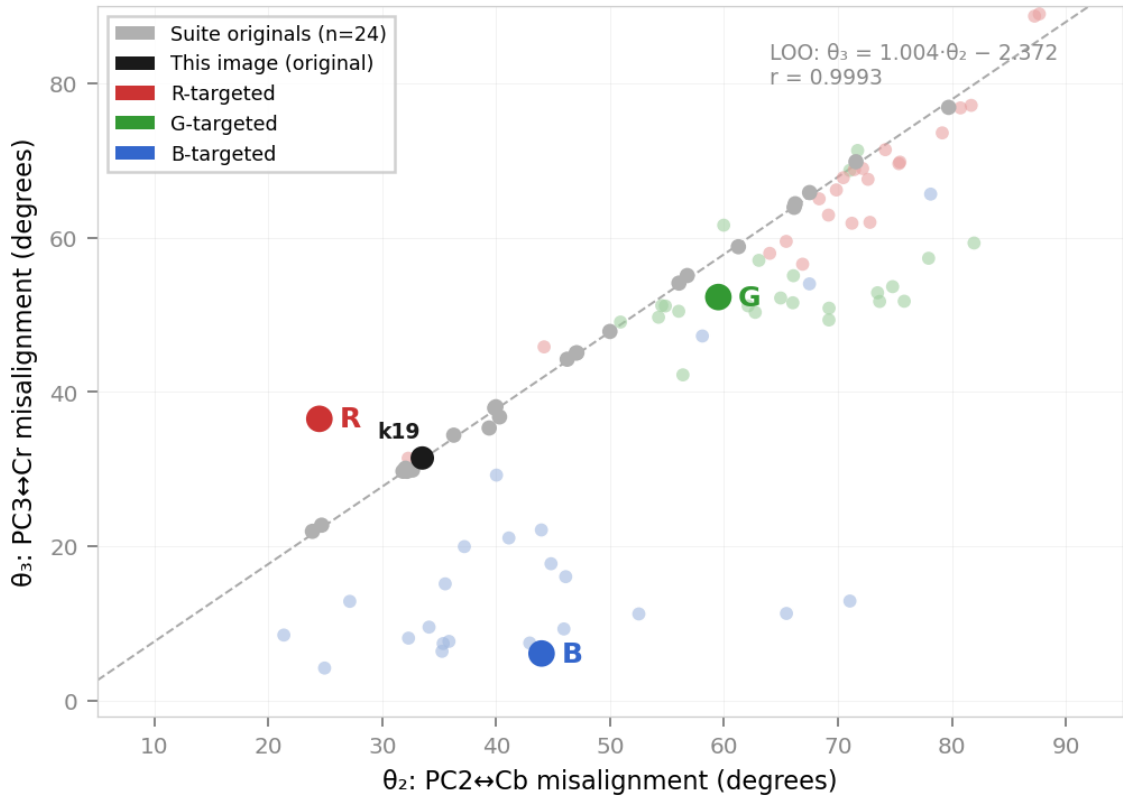


Compression Profile: KODIM19 512x768 | 24-bit RGB | PNG (lossless)

lighthouse in Maine | Strongly One-Dimensional (PC1 = 93.36%)

1. Geometric Displacement Under Directional Perturbation



2. Pairwise Correlation and Angular Misalignment

Version	R-G	R-B	G-B	Avg r	theta2	theta3	L00 dev
Original	0.9668	0.8155	0.9151	0.899	33.51°	31.45°	+0.18°
R-targeted	0.0392	-0.0079	0.9156	0.321	24.47°	36.53°	+14.33°
G-targeted	0.0588	0.9139	0.1326	0.368	59.5°	52.32°	-5.05°
B-targeted	0.9790	-0.0725	-0.1393	0.397	43.98°	6.1°	-35.68°

R-targeted isolates red (R-G, R-B < 0.04) while preserving G-B coupling at 0.916.
G-targeted isolates green (R-G, G-B < 0.13) while preserving R-B at 0.914.
B-targeted preserves R-G at 0.979 while driving R-B and G-B near zero.
All three perturbation directions depart from the L00 constraint manifold ($r = 0.9993$).

Compression Profile: KODIM19 512x768 | 24-bit RGB | PNG (lossless)

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3. File Format and Size Comparison

All versions depict the same scene. BPP = (file size in bits) / (pixel count).

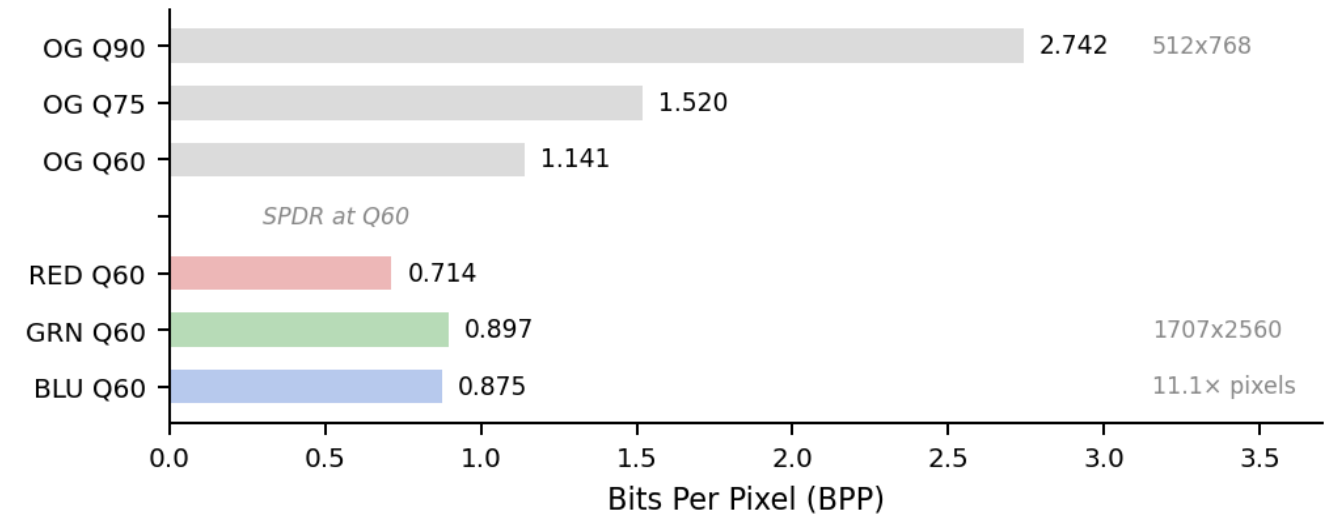
ORIGINAL (kodim19)

Version	Format	Resolution	Size	BPP	Pixels
PNG (lossless)	PNG	512x768	655.7 KB	13.661	393,216
JPEG Q90	JPEG	512x768	131.6 KB	2.742	393,216
JPEG Q75	JPEG	512x768	73.0 KB	1.520	393,216
JPEG Q60	JPEG	512x768	54.8 KB	1.141	393,216

REDISTRIBUTED (kodim19 SPDR at Q60)

Version	Format	Resolution	Size	BPP	Pixels
RED Q60	JPEG	1707x2560	380.6 KB	0.714	4,369,920
GRN Q60	JPEG	1707x2560	478.7 KB	0.897	4,369,920
BLU Q60	JPEG	1707x2560	466.9 KB	0.875	4,369,920

4. Quality Ladder: SPDR Q60 vs Original Q90



All three perturbation directions at Q60 produced lower BPP than the original at Q90, while carrying 11.1× the original pixel count.

Compression Profile: KODIM19 512x768 | 24-bit RGB | PNG (lossless)

lighthouse in Maine | Strongly One-Dimensional (PC1 = 93.36%)

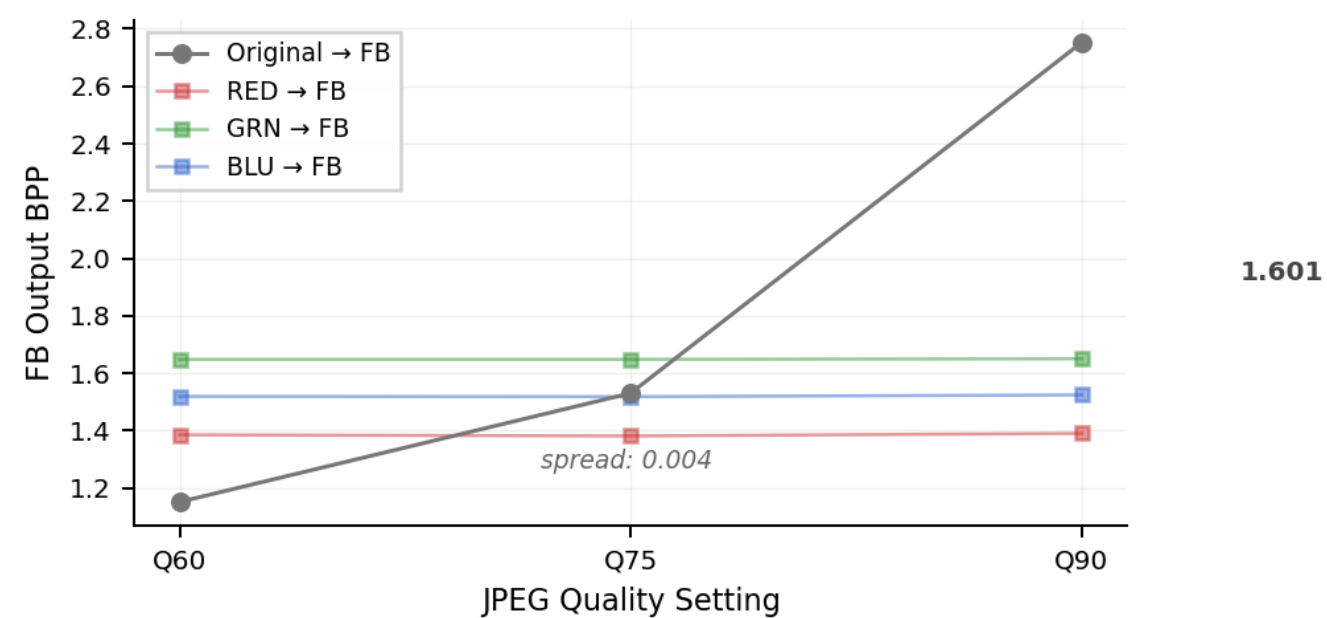
5. Third-Party Compression Pipeline (Facebook/Meta)

Each version uploaded to Facebook. Platform-compressed output measured at steady state (FB2).

Pipeline	Input Res	Output Res	FB BPP	BPP reduction
OG PNG → FB	512x768	512x768	2.572	—
R-targeted → FB	1707x2560	1366x2048	1.390	46.0%
G-targeted → FB	1707x2560	1366x2048	1.650	35.8%
B-targeted → FB	1707x2560	1366x2048	1.524	40.7%
ABC (no perturbation) → FB	1707x2560	1366x2048	1.415	45.0%

ABC control: same pipeline, same resolution, no covariance perturbation. Avg |r| unchanged (0.8973 vs 0.899 original). Comparable BPP reduction was not observed in the ABC control.

6. Quality Parameter Sensitivity Through Facebook



Original: Q60 → Q90 produces a 1.601 BPP spread through Facebook.
Redistributed: Q60 → Q90 produces a 0.004 BPP spread through Facebook.
Redistributed versions produced substantially smaller Q60-90 BPP variation through the measured pipeline.

7. Pipeline Summary – kodim19

Best BPP (Q60, direct)	0.714	1707x2560, 4,369,920 px
Best BPP through FB	1.390	1366x2048 output
vs OG Q90 → FB	49.5%	SPDR Q60→FB vs OG Q90→FB
Pixel ratio	11.1×	4,369,920 vs 393,216